MOHAMED AZIZ SAKISS

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SUMMARY

AI & Machine Learning Engineer specializing in Computer Vision, Generative AI, LLMs and Automations. Experienced in building production-ready pipelines, real-time inference models, and integrating Blockchain for data integrity. Proven track record in implementing state-of-the-art (SOTA) research—such as YOLOv11 and Transformers—to bridge the gap between high-accuracy AI and Cybersecurity. Passionate about deploying scalable, high-impact solutions for international startups and enterprise-level infrastructure.

EDUCATION

EPI International Multidisciplinary School — Sousse, Tunisia
Engineering Degree in Artificial Intelligence & Data Science — *Sep 2021 – Aug 2026 (Expected)*

TECHNICAL SKILLS

Generative AI & LLMs: LLMs, LangGraph, Agents, RAG, Prompt Engineering, Embeddings, LangChain, Hallucination Analysis
Machine Learning: Transformers, CNNs, Scikit-Learn, XGBoost, Random Forest, TensorFlow/Keras, PyTorch
Computer Vision: YOLO (v8/v11), OpenCV, MediaPipe, ResNet, Image Classification, SAHI, Object Detection
Backend, MLOps & Systems: FastAPI, Node.js, PostgreSQL, Apache Spark (PySpark), Docker, CI/CD, GitHub Actions, Git, Streamlit, Blockchain
Programming: Python, C++, JavaScript/TypeScript, Bash, SQL

PROFESSIONAL EXPERIENCE

AI Engineering Intern — Tunis, Tunisia
Tunisie Telecom — *Feb 2026 – Jul 2026*

- **Real-Time Vision & ML Pipeline:** Engineered ErgoFit to monitor live ergonomics, combining MediaPipe pose landmark extraction with a Random Forest classifier to achieve 98% posture detection accuracy.
- **Hybrid RAG & Hallucination Prevention:** Developed "ErgoCoach," an LLM assistant that fuses structured PostgreSQL data (user error patterns) with certified ergonomic PDFs to generate medically grounded advice while strictly preventing model hallucinations.
- **Autonomous Reporting Agent:** Designed a scheduled AI agent that analyzes temporal database metrics to generate and dispatch personalized weekly reports, offering actionable insights for continuous posture improvement.
- **Architecture & Live Deployment:** Built a hybrid backend (Node.js/FastAPI), containerized microservices using Docker, and automated CI/CD pipelines via GitHub Actions to launch the platform live at ergofit.live.

Software Development Intern — Remote, Switzerland
Business Scalable — *Jul 2025 – Aug 2025*

- **Production-Ready RAG Agents:** Designed and deployed NLP-based chatbots using LLMs (Gemini/OpenAI) and LangChain to assist users with instant data retrieval and troubleshooting. This delivered clear business value by automating customer support interactions, significantly reducing manual workload, and slashing response latency.
- **End-to-End Pipeline Engineering:** Built scalable data pipelines for business applications, integrating REST APIs with backend systems to ensure real-time data processing.
- **Documentation & Deployment:** Collaborated with the engineering team to document model performance metrics and streamline the deployment process using Docker and CI/CD principles.

PROJECTS

JobPilot: Autonomous Agentic SaaS | ***Python, LangGraph, FastAPI, Whisper*** — Aug 2026 – Present

- **Core Product & Business Value:** Engineering an autonomous agentic SaaS platform that intelligently parses resume data to automate targeted job applications. This drastically reduces manual job-hunting time for users and scales their application volume to significantly enhance their hiring chances.

- **Real-Time Conversational Voice Agent:** Engineering a fully verbal, audio-in/audio-out mock interview AI. The system actively listens to the user's spoken answers via OpenAI Whisper, processes the context against their resume and target job descriptions, and responds dynamically using Text-to-Speech (TTS) to simulate a natural, human-like interviewer.
- **Backend Orchestration:** Architecting complex multi-agent workflows using LangGraph to manage application logic, coupled with a high-performance FastAPI backend to serve the interactive dashboard and handle asynchronous tasks.

Secure Brain Tumor Classification (NeuroChain) | ***Python, TF, Blockchain*** Nov 2025 – Jan 2026

- Developed an end-to-end medical diagnostic application merging Deep Learning with a custom Blockchain backend to ensure immutable patient data logging.
- Blockchain Integration: Engineered "NeuroChain," a Python-based ledger using SHA-256 cryptography. Implemented a strict verification protocol that flags unauthorized modifications with a "Tampering Detected" alert.
- Interactive Dashboard: Built a full-stack Streamlit interface allowing doctors to upload MRI scans, view real-time classification results, and verify cryptographic integrity.
- Model Performance: Trained a high-precision CNN model (TensorFlow/Keras) for tumor detection.

Full-Stack Generative AI Agent for E-Commerce | ***Python, Gemini, RAG*** Nov 2025 – Dec 2025

- Designed and deployed a tool-using Generative AI agent powered by Gemini Flash 2.5 to enable real-time inventory queries and order tracking.
- Implemented RAG (Retrieval-Augmented Generation) to ground responses in structured ERP data, significantly reducing hallucinations and improving response accuracy.

Real-Time Object Detection for Autonomous Driving | ***YOLOv11, PyTorch*** Oct 2025 – Dec 2025

- Implemented and fine-tuned YOLOv11m on the BDD100K dataset (7,000+ images) for autonomous vehicle perception.
- Conducted ablation studies to optimize mAP vs. inference latency for edge deployment and improved small-object recall using SAHI.

Real-Time Credit Card Fraud Detection System | ***PySpark, MLlib*** Nov 2025 – Dec 2025

- Built an end-to-end Big Data pipeline processing 1M+ transactions using Apache Spark.
- Solved extreme class imbalance using custom class weighting (5.7x penalty), achieving 98.8% recall and AUC $>$ 0.99 with a weighted GBT model.

CERTIFICATIONS

Machine Learning Specialization — DeepLearning.AI (Mar 2026)

Generative AI: Prompt Engineering Basics — IBM (Mar 2026)

Building RAG Agents with LLMs — NVIDIA (Nov 2025)

Fundamentals of Deep Learning — NVIDIA (Nov 2025)

Building Transformer-Based Natural Language Processing Applications — NVIDIA (Feb 2026)

Foundations of Data Science — Google (Jan 2026)

LANGUAGES

Arabic: Native | **English**: Professional | **French**: Professional | **German**: A2 (Basic)